

8.ANALYSIS OF INCOME TAXES

Differences between the balance sheet carrying value and the tax base of an asset may be permanent or temporary.

tax expense = Tax Payable + Δ DTL - Δ DTA + Δ Allowance

Net Income = Pre-Tax Income - Tax Expense = Pre-Tax Income - (Tax Payable + Δ DTL - Δ DTA + Δ Allowance for DTA)

➤ From the financial reporting—财务报表	Balance sheet approach	Income statement approach
Carrying value/carrying amount Balance sheet value of an asset or liability	Step 1: Identify Carrying value and Tax base for every asset and liability item on B/S	Step 1: Calculate Accounting profit & Taxable income
➤ From the tax reporting—税务报表	Step 2: Calculate the difference two bases For assets = Carrying value - Tax base For liabilities = Tax base - Carrying value <i>Positive figure * tax rate = DTL</i> <i>Negative figure * tax rate = DTA</i>	Step 2: Calculate Income tax expense & Tax payable Δ DTA = (Taxable income (较大) - Accounting profit) * tax rate Δ DTL = (Accounting profit (较大) - Taxable income) * tax rate
Tax base Net amount of an asset or liability used for tax reporting purposes.		注：是DTA还是DTL看第一年的数值谁较大

(Carry amount - tax base) x tax rate = DTA/DTL

(Tax Income - Accounting Profit) x tax rate = DTA/DTL

The changes in deferred tax assets and liabilities are added to income tax payable to determine the company's income tax expense (or credit) as it is reported on the income statement.

If the company later realizes it **won't earn enough taxable income** to use all the DTA, it **writes down (reduces)** the DTA through a **valuation allowance** to reflect the portion it likely won't recover.

DTL can be treated as Equity or Liability

If DTL is reversable in foreseeing future, DTL is treated as liability.

Only temporary differences create deferred tax assets or liabilities. If the DTL is unlikely to reverse (ex: increase its investment in depreciable asset), it doesn't represent a future cash outflow, aligning it more closely with equity than with a liability.

Treating the full DTL as equity avoids overestimating liabilities in the firm's capital structure.

DTA, DTL scenario:

A **DTL** arises from acceleration depreciation because accelerated depreciation reduces taxable

income faster in the early years, deferring taxes to later years when taxable income increases. DTL will approach to 0.

Restructuring generates a DTA because the costs are recognized for financial reporting when the restructuring is announced, but they are not deducted for tax until actually paid

In the United States, firms that use LIFO for their financial statements are required to use LIFO for tax purposes, so no temporary differences result. However, in countries where this is not a requirement, temporary differences can result from the choice of inventory cost low method

A deferred tax adjustment is made to stockholders' equity to reflect the future tax impact of unrealized gains or losses on available-for-sale marketable securities (OCI)

Tax rate

The footnote disclosure in the financial statements must reconcile the statutory tax rate to the effective tax rate

Statutory tax rate: corporate income tax rate in which the company is domiciled

effective tax rate: $\text{income tax} / \text{pretax income}$ (Theory paid)

cash tax rate: $\text{cash tax paid} / \text{pretax income}$ (actually paid)

The effective rate is most useful for forecasting future earnings; however, it may need to be adjusted to remove the impact of transitory components. The statutory rate is the tax rate where the company is domiciled for reporting purposes. The cash tax rate is most applicable when forecasting future tax payments and cash flows.

Differences between accounting profit and taxable income

1. Which of the following tax definitions is least accurate?
 - A. Taxable income is income based on the rules of the tax authorities.
 - B. Taxes payable are the amount due to the government.
 - C. Income tax expense equals pretax income times the statutory tax rate.

ans: C

Pretax income and income tax expense are not always linked because of temporary and permanent differences.

If the tax base of an asset is less than the carrying value of the asset and the difference is not expected to reverse in the future, this will result in:

- A. the creation of a DTL.
- B. the creation of a DTA.
- C. neither the creation of a DTA nor a DTL.

ans: C

Timing differences that do not reverse, referred to as permanent differences, do not result in the creation of DTAs or DTLs. If the timing difference was expected to reverse, a DTL would be created because carrying value > tax base

Which of the following is least likely to result in the creation of a deferred tax asset (DTA)?

- A. Expenses or losses that are tax deductible before they are recognized in the income statement.
- B. Revenues or gains that are taxable before they are recognized in the income statement.
- C. Tax loss carryforwards that are available to reduce future taxable income

ans: A

Expenses that are tax deductible before they are recognized in the income statement result in deferred tax liabilities. For example, accelerated depreciation in the tax returns versus straight-line depreciation in the accounts results in more depreciation being recognized in the tax returns, in early periods, and results in carrying value exceeding the tax base. If revenues or gains are taxable before they are recognized in the income statement, DTAs are created. Unearned revenue is a good example of this. Tax loss carryforwards result from negative taxable income. Rather than triggering the tax authorities to pay tax to the firm, they are deferred and can be offset against future taxable income. As a result, the firm will pay less tax in the future, and a DTA is created.

Roadrunner, Inc., reported the following information in its tax returns and financial statements:

Which of the following is closest to the tax expense reported in Roadrunner's 20X7 income statement?

Tax return extract:	\$
Tax payable for the year ended 20X7	500,000
Balance sheet extracts:	
DTA year-end 20X7	300,000
DTA year-end 20X6	350,000
DTL year-end 20X7	500,000
DTL year-end 20X6	400,000

- A. \$500,000.
- B. \$550,000.
- C. \$650,000

ans: C

Change in DTA: $\$300,000 - \$350,000 = -\$50,000$

Change in DTL: $\$500,000 - \$400,000 = \$100,000$

Tax expense = $\$500,000 + \$100,000 - (-\$50,000) = \$650,000$

Dta & dtl

Enigma KK uses straight-line depreciation in its accounts, while using accelerated depreciation for tax purposes. For both financial and tax reporting, the company assumes a five-year useful economic life and zero residual value. In the current year, accounting depreciation is ¥10,000,000 and tax allowable depreciation is ¥8,000,000. Which of the following statements relating to the impact of depreciation in the current year is most accurate?

- A. The DTL will increase.
- B. The tax expense will be greater than tax payable.
- C. The carrying value of the asset is greater than the tax base

ans: C

The carrying value is higher than the tax base of the asset due to the use of accelerated depreciation in the tax returns versus straight-line in the accounts. The tax allowable depreciation is greater than the accounting depreciation, indicating that the temporary timing difference is reversing. As tax allowable depreciation is now lower than accounting depreciation, the DTL will begin to decline. Decreases in the DTL reduce the tax expense. The difference between carrying value and tax base will decrease as the timing difference reverses until, at the end of the asset's life, they both equal zero.

A U.S. GAAP reporting firm reports an increased valuation allowance at the end of the current period. What effect will this have on the firm's income tax expense in the current period?

- A. Increase.
- B. Decrease.
- C. No effect.

ans: A

Recognizing a greater valuation allowance reduces the net value of a DTA, which increases income tax expense in the current period.

An analyst is comparing a firm to its competitors. The firm has a DTL that results from accelerated depreciation for tax purposes. The firm is expected to continue increasing its purchases of PP&E in the foreseeable future. How should the liability be treated for analysis purposes?

- A. It should be treated as equity at its full value.
- B. It should be treated as a liability at its full value.
- C. The present value should be treated as a liability, with the remainder being treated as equity

ans: A

The DTL is not expected to reverse in the foreseeable future because the firm is expected to continue to increase its investment in depreciable assets, and accelerated depreciation for tax on the newly acquired assets delays the reversal of the DTL. The liability should be treated as equity at its full value